

Assembly Line — Operation & Reference Manual

Equipment Manual · Machine Type: Assembly Line

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Owner	Manufacturing Engineering
Applies To	Assembly Line — Plant B (Chennai), Plant C (Ahmedabad); units Assembly_Line_1 through Assembly_Line_20

1. Overview

Semi-automated assembly stations combining conveyor transport, pick-and-place fixtures, and manual fitment stations. Receives milled and pressed components from the CNC and Press lines.

2. Machine Specifications

Parameter	Specification
Conveyor speed	0.2 – 0.6 m/s, variable
Station count	6 fitment stations per line
Ideal cycle time	1.0 – 1.4 min/part
Torque tools	Calibrated pneumatic torque wrenches, ±3% accuracy

3. Startup Procedure

- 1 Verify all station e-stops are reset and conveyor guarding is in place.
- 2 Confirm torque tools are within current Calibration_SOP validity window before shift start.
- 3 Start conveyor at crawl speed and verify tracking across all stations.
- 4 Ramp to production speed once first 5 units pass visual quality check.

4. Common Downtime Codes on This Machine Type

Code	Description	Severity	Reference SOP
DT-01	Hydraulic System Failure	Major	Hydraulic_Maintenance_SOP
DT-04	Sensor / Limit Switch Fault	Minor	Electrical_Troubleshooting_SOP
DT-05	Conveyor Jam / Misfeed	Minor	Inspection_Checklist
DT-07	Operator / Shift Handover Delay	Minor	Plant_Maintenance_Policy

■ Keep loose clothing and hands clear of conveyor pinch points at all times; guarding must never be defeated.